BIOT Module 6

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1. Introduction

Imagine two people who take the same genetic test for Alzheimer's disease and get the same result back: "higher than average risk." It is natural to assume the number means the same thing for both of them. Whether it does depends on something the report never mentions: who the test was built on.

Tests like this are called polygenic risk scores. They add up many small differences in a person's DNA, each one weighted by how strongly it was linked to the disease in earlier studies. Those studies have mostly been done in people of European ancestry, about four in five participants by one count (Martin et al., 2019). A score learned in one population is then applied to everyone.

In Module 1 I looked at APOE, the gene with the largest single effect on Alzheimer's risk. Here APOE comes back, this time as one ingredient in a 22-SNP score, applied to people from five ancestry groups.

This brings us to the main question of this module: does a published Alzheimer's risk score work equally well for people of different ancestries, and if it does not, is that because of the biology or because of the data?

2. Disease and data choice, with biological interpretation

I stayed with Alzheimer's disease, the disease of Modules 1 to 4, so that the APOE work from Module 1 carries through. The data is the course cohort from Canvas, used unchanged. It has three parts:

Part	What it is	Source
The score	22 SNPs, each with a published weight, including APOE e4 (rs429358) and e2 (rs7412)	PGS Catalog PGS000334 (Zhang et al., 2020)
The people	2,336 real genomes from five ancestry groups	1000 Genomes phase 3 (The 1000 Genomes Project Consortium, 2015)
Disease status and age	case or control, age	simulated for the course

The five groups are European (EUR, 500 people), African (AFR, 500), East Asian (EAS, 500), South Asian (SAS, 489) and Admixed American (AMR, 347).

The first two parts are real. The third is not, because 1000 Genomes collects DNA but no medical records. To make a disease label, the course linked each SNP in the score to a hidden "causal" SNP nearby, chosen to travel closely with it in Europeans, and drew case status from those causal SNPs plus age and chance. The top 30% became cases, by the same rule in every group. So the scores and their spread across ancestries are real measurements, while every AUC and calibration result is an illustration of a real effect, not a measurement of real Alzheimer's risk.

Why ancestry matters biologically: two things differ between populations. The first is how common each variant is. APOE e4, the score's largest weight, has an allele frequency of 0.26 in the African-ancestry group here, 0.15 in Europeans and 0.09 in East Asians, in line with published figures. The second is linkage disequilibrium (LD), how tightly neighbouring variants are inherited together. African-ancestry genomes carry the oldest human variation and have shorter LD blocks, so a SNP that stands in well for a causal variant in Europeans can be a poor stand-in elsewhere.

3. Method

1. *Harmonise alleles:* match each score SNP to the genotype's REF and ALT; keep, flip (2 − dosage) or drop. Result: 22 used, 13 flipped, 0 dropped.
2. *Compute the PRS:* dosages times weights, summed per person, standardised across the cohort.
3. *One cutoff:* take the pooled top-10% value and count the share of each group above it.
4. *Discrimination:* AUC overall and within each ancestry.
5. *Calibration:* fit a logistic model on Europeans only, apply it to everyone, compare predicted with observed risk.
6. *Clinical value:* 5-fold cross-validated AUC, age alone against age plus PRS.
7. *Extra check (`scripts/extra_checks.py`):* each group's AUC without the two APOE SNPs, and with e4 count alone.

4. Results

Table 1: Results by ancestry. Cross-validated AUC: age alone 0.562, age plus PRS 0.618.

	EUR	SAS	EAS	AMR	AFR	All
Mean PRS (SD)	+0.04	−0.25	−0.47	−0.10	+0.74	
% above top-10% cutoff	7.8	4.3	2.2	6.3	28.0	10
AUC	0.672	0.611	0.575	0.555	0.493	0.599
Predicted risk (EUR model)	0.314	0.273	0.246	0.293	0.418	
Observed case rate	0.314	0.256	0.280	0.265	0.374	0.300

Three things stand out. The score works best in the group it was built in: AUC 0.672 in Europeans, falling to 0.493, which is chance, in the African-ancestry group. The African-ancestry group also scores highest on average (+0.74 SD), mostly because e4 is more common there. And APOE carries much of the score: everyone in the pooled top 10% has at least one e4 allele, and e4 alone gives 0.645 of the European AUC of 0.672.

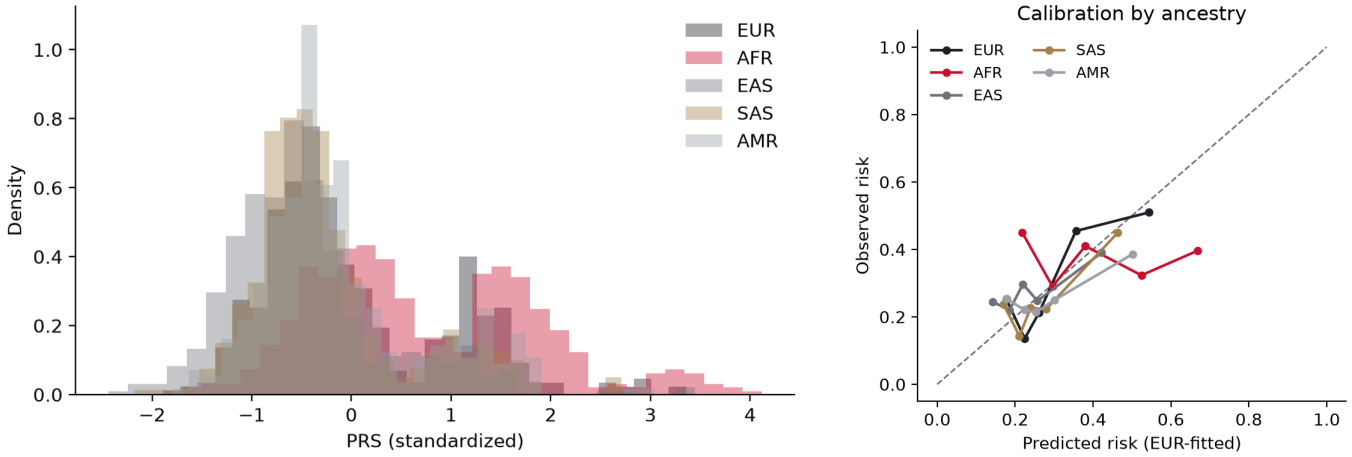


Figure 1: Left: PRS distribution by ancestry. The upper bump is people carrying APOE e4, by far the largest weight in the score, and the African-ancestry group sits furthest to the right. Right: calibration of the EUR-fitted model in each group; points on the dashed line are honest predictions.

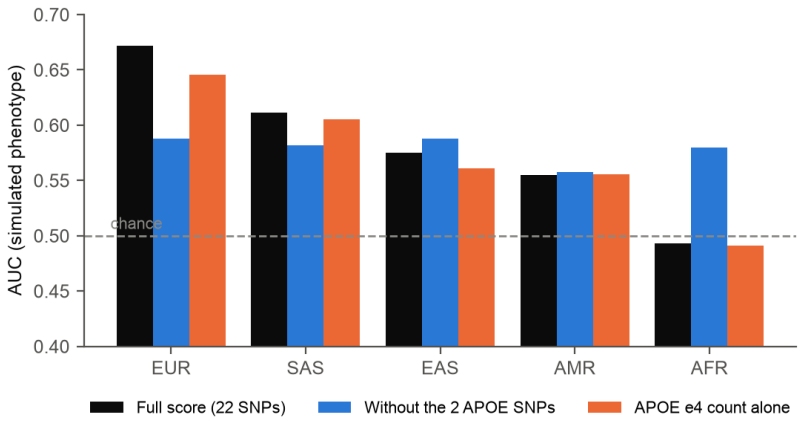


Figure 2: AUC by ancestry for the full score, the score without the two APOE SNPs, and APOE e4 count alone (my extra check). Without APOE, every group sits near 0.58.

5. Interpretation

1. Portability: the 22 SNPs are not the causal variants; they are markers found in European studies that travel with the causal variants because of LD in Europeans. Elsewhere the LD is different: the mean r^2 between marker and causal SNP is 0.82 in EUR, about 0.73 to 0.75 in EAS, SAS and AMR, and 0.44 in AFR (course data README). The AUC follows: highest in EUR (0.672), lowest in AFR (0.493). Allele frequency differences (e4 is 0.26 in AFR, 0.09 in EAS) also shift whole groups up or down, regardless of risk.

2. One threshold, many groups: no. A pooled top-10% cutoff flags 28.0% of African-ancestry people but only 2.2% of East Asians, while the real case rates (37% and 28%) are not nearly that far apart. Every flagged person carries e4, so the cutoff is largely an e4 test. If a clinic applied it unchanged, it would send many African-ancestry patients for follow-up on a signal that does not predict their outcome here (AUC 0.49), and would miss most high-risk East and South Asian patients. Each group would need its own cutoff, or a score adjusted for ancestry.

3. Discrimination against calibration: discrimination asks whether cases rank above controls; calibration asks whether a predicted 40% means 40%. In EUR both look fine: AUC 0.672, and mean predicted risk equals the observed rate (0.314). In AFR they disagree: the ranking is at chance (0.493), yet the model predicts the highest risk of any group (0.418 against 0.374 observed), and its top bin promises about 0.67 where about 0.40 are cases. EAS shows the reverse: fair ranking (0.575) but risk underestimated (0.246 against 0.280).

4. Clinical utility and limits: the PRS adds to age (cross-validated AUC 0.562 to 0.618), but 0.62 is far from a diagnostic test. It could help research, for example to enrich prevention trials with higher-risk Europeans. It should not be used to tell an individual their risk, and not in non-European groups until checked there. Before deployment I would need a real cohort with diagnosed AD in every ancestry where it would be used (for example All of Us or the Alzheimer's Disease Sequencing Project), calibration within each group, and evidence that it improves decisions beyond age and family history. The phenotype here is simulated, so none of these numbers measure real AD risk.

5. Artifact against biology: the near-chance AUC in AFR comes from how the phenotype was built. The simulation put e4's effect on a nearby proxy SNP that travels with e4 in Europeans, but in Africans that proxy has r^2 of about 0.01 with e4. So here African e4 carriers get no extra disease, and e4, the score's biggest weight, becomes noise. My extra check confirms it: without the two APOE SNPs the AFR AUC rises from 0.493 to 0.580, level with other groups, and e4 alone scores 0.491. In real cohorts e4 is a risk allele in African-ancestry people too, though weaker than in Europeans, partly depending on the local ancestry around APOE (Rajabli et al., 2018). A score failing in a group because of the test data does not mean the gene is unimportant there.

6. Discordance and limitations

Where the results disagree: AFR has the highest mean score, the highest case rate and the highest predicted risk, but the score cannot rank cases there. EAS ranks reasonably but is under-predicted. High scores and high predicted risk are not the same as a working test.

What I cannot claim:

1. Anything about real AD risk. The phenotype is simulated, with the noise tuned so EUR AUC is about 0.67.
2. That AFR results reflect real biology. As explained in question 5, the APOE result is built into the simulation.
3. That these group results hold for individuals. Groups are broad: AFR and AMR include admixed populations (for example ASW, ACB, MXL and PUR), whose LD and e4 effects vary person to person.
4. That 22 SNPs represent modern scores. Larger scores with thousands of SNPs behave differently, though they usually lose even more accuracy outside Europe (Martin et al., 2019).
5. Precise numbers. Group sizes are 347 to 500, so each AUC is uncertain by a few hundredths.

If I did this again, I would: fit and calibrate the model within each ancestry, try an ancestry-adjusted cutoff, report confidence intervals for each AUC, and test the score with and without APOE in a real cohort with diagnosed AD.

Proof of real data

The fingerprint printed by Section 2 of the notebook; the checksums match the ones listed in the course data README.

```
source      : REAL
people      : 2336
score size  : 22 variants
by ancestry : {'AFR': 500, 'EAS': 500, 'EUR': 500, 'SAS': 489, 'AMR': 347}
  genotypes.csv: sha256 9cf41a1f
  variants.csv: sha256 f7a87d3f
  score.csv: sha256 7f80e8ff
  meta.csv: sha256 6cc55b6e
```

References

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Data: PGS Catalog score PGS000334, https://www.pgscatalog.org/score/PGS000334. 1000 Genomes phase 3, https://www.internationalgenome.org.